vTrig B
Evaluation Kit



Hardware spec

Revision 2

Vayyar Imaging Ltd.

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Notice

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Revision History

Revision	Date	Author	Description
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Chapter 1
Introduction

This guide provides information on the **Vayyar vTrig Evaluation Kit.** vTrig is a three-dimensional, RF-based sensor and imaging system that is installed on a wall or ceiling, and detects objects in a defined arena and displays a real-time image of their positions within the arena.

1.1 Audience

This document is targeted at the following audiences:

- Engineering and technical marketing personnel who are interested in evaluating Vayyar 3D imaging capabilities and are considering integrating Vayyar solutions into their products.

1.2 Release Information

This is a first Hardware version for demonstration of system capabilities. Both algorithmic capabilities and features are under continuous development and improvement.

1.3 FCC Notice

This kit is designed to allow:

- Product developers to evaluate electronic components, circuitry, or software associated with the kit to determine whether to incorporate such items in a finished product.
- Software developers to write software applications for use with the end product. This kit is not a finished product and when assembled may not be resold or otherwise marketed unless all required FCC equipment authorizations are first obtained. Operation is subject to the condition that this product not cause harmful interference to licensed radio stations and that this product accept harmful interference. Unless the assembled kit is designed to operate under part 15, part 18 or part 95 of this chapter, the operator of the kit must operate under the authority of an FCC license holder or must secure an experimental authorization under part 5 of this chapter.

1.4 Glossary

Table 1: Glossary

Term	Definition
AC	Alternating Current
API	Application Programming Interface
DC	Direct Current
DLL	Dynamic Link Library
EVK	Evaluation Kit
GUI	Graphical User Interface
IP	Internet Protocol. An IP address is a numerical identifier assigned to a computing device or node in a TCP/IP network. The address is used to locate and identify the node in communications with other nodes on the network.
MCU	Micro-Controller Unit
PCB	Printed Circuit Board
RF	Radio Frequency
SoC	System-on-chip

Chapter 2
System Overview

The vTrig system is designed to demonstrate Vayyar's 3D imaging non-contact RF sensors, which enable to identify multiple objects (“blobs”) in a defined space (“*arena*”).

2.1 System Architecture

The vTrig system architecture and connectivity scheme is illustrated in the figure below.

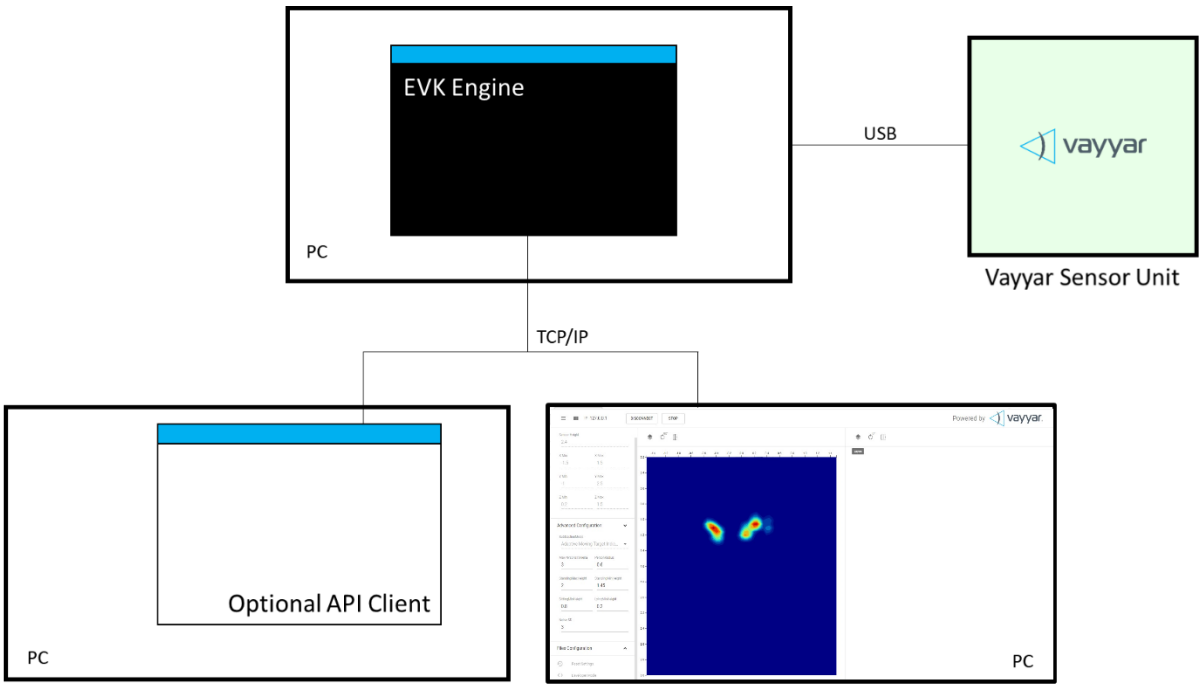


Figure 1: vTrig System Architecture

The vTrig system is comprised of the following components:

- **The vTrig RF Sensor Unit.**
- **The EVK Engine.** Host software that performs signal processing functions. The software runs in a command window in a PC, which is connected to the vTrig Sensor via a USB cable.
- **Web GUI.** User-interface client software that communicates with the EVK Engine via a TCP/IP connection. The Web GUI client is supported by the following browsers:
  - Google Chrome
  - Mozilla Firefox
- **API Client.** Optional, customer-developed software that communicates with the EVK Engine via a TCP/IP connection.

NOTE

The Web GUI and API Client software can run either on their own dedicated PC, or on the same PC as the EVK Engine.

2.2 Hardware

The figure below presents a high-level block diagram of the PCB that houses the EVK sensor unit:

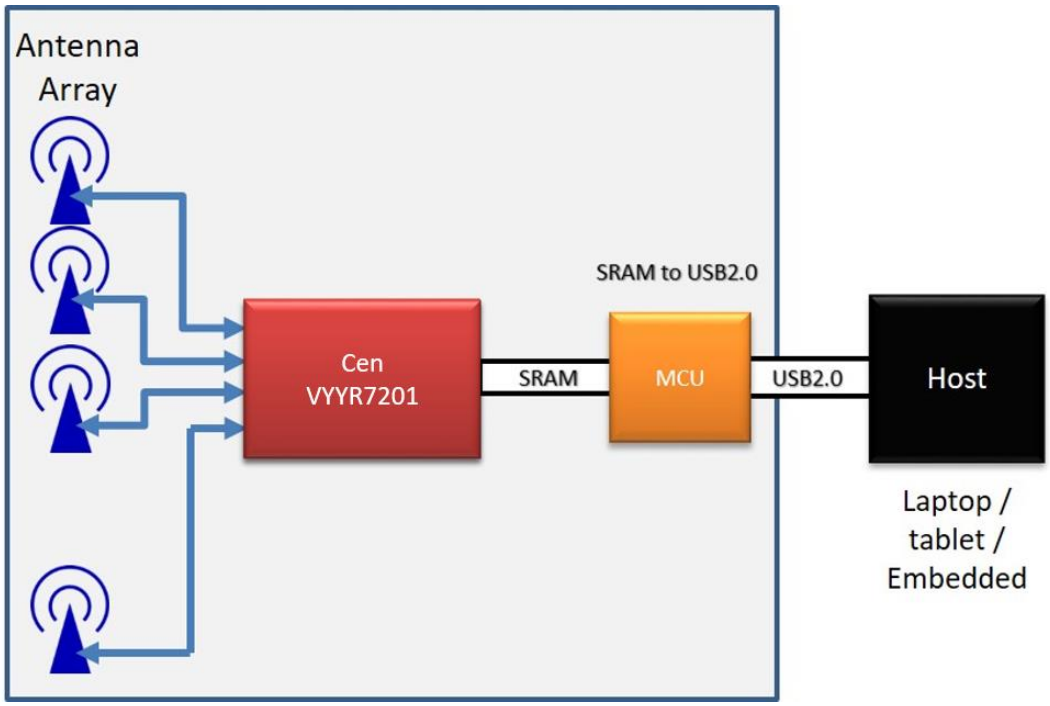


Figure 2: High-Level Block Diagram

The following components are embedded in the sensor unit PCB:

- Vayyar VYYR7201-B0 system-on-chip (SoC).
- 40 embedded Tx/Rx linear polarized PCB embedded, wide-band antennas.
- A digital interface that communicates with a host PC via a USB-2.0 connection. The interface is implemented using on-board Cypress micro-controller unit (MCU).
- A Crystal used as a clocking source for the Cypress MCU.

  A TCXO (Temperature Compensated Crystal Oscillator) used as a clocking source for the VYYR7201-B0 SoC.
- Power supply system supporting 3.3V, 2.5V, 1.27V and 1.2V DC/DC converters.
- Non Volatile Memory (NVM) used to store inventory and calibration information.

2.2.1 Sensor Unit Specification

Table 2: Sensor Unit Specification

Parameter	Specification	Notes
Transceivers	40 Tx/Rx	
Frequency Band	60 - 64 GHz	Configured by software
Field of View (FoV)	Refer to section 2.2.5 Antenna and RF Characteristics	
Range Resolution	$\frac{C}{2BW} = \frac{3e^8}{2 \cdot 4e^9} = 3.75\text{cm}$	Depends on the configured BW
Range Accuracy	< < Range Resolution	Depends on the target strength and shape
Angular Resolution	$\Delta\theta \approx \frac{\lambda}{D} \approx \frac{3e^8}{64e^9 \cdot 0.04} \approx 0.117rad \approx 6.7deg$	λ – Wavelength [@64GHz] D – Length of the array [0.04m]
Angular Accuracy	<< Angular Resolution	Depends on the target strength and shape
Dimensions	Without case: 80 mm x 80 mm With case: 105 mm x 85 mm	Dimensions refer to vTrig board, that support lots of Vayyar internal debug options and are not indicative to a final product size

2.2.2 Digital Interface

The digital interface connects the host PC with the EVK board. The interface is implemented on the EVK board using an MCU and a micro-USB connector.

Micro-Controller Unit (MCU)

Cypress's EZ-USB FX3™ MCU integrates the USB 2.0 physical layers (PHYs), and contains a 32-bit ARM926EJ-S microprocessor for powerful data processing and for building custom USB SuperSpeed applications. The integrated processor contains a unit called General Programmable Interface (GPIF), which provides easy connectivity to popular interfaces.

USB Connector

The onboard micro-USB-B 2.0 connector is used for power supply and for data transfer between the sensor unit and the host PC. The connector supports the USB 2.0 protocol in High-Speed Mode (480 Mbps).



Figure 3: vTrig Connectivity

2.2.3 RF Interface - Antennas

The array of 40 antennas serves to connect the sensor unit with the environment. The antennas are embedded in the PCB, as depicted in the figure below.

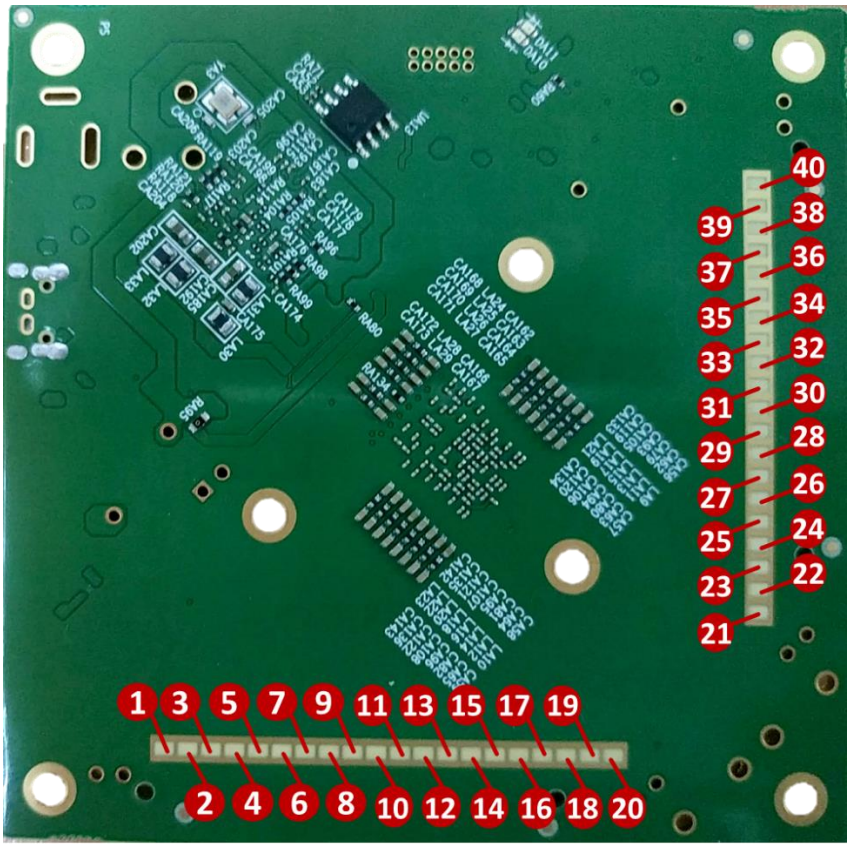


Figure 4: Antenna Positions and Numbering

For each transmitting (TX) antenna, there are multiple receiving (RX) antennas for collecting and recording the received RF signals. Each RX antenna and its associated TX antenna serve as an *antenna pair*.

2.2.4 Electrical Specifications

The following electrical specifications are for the sensor unit. The current consumption specifications are based on lab measurements performed by Vayyar.

Table 3: Sensor Unit Electrical Specification

Parameter	Value
Vcc supply voltage	5 VDC
Max current consumption	~1.3 A
Average current consumption	~1 A

2.2.5 Antenna and RF Characteristics

The following are antenna and RF characteristics, based on the simulation of a single antenna over an infinite ground plane:

Table 4: Antenna and RF Characteristics

Parameter	Value	Comment
Gain	5.8 dBi @64 GHz	At 0˚ Elevation, 0˚ Azimuth
Antenna -3 db Beamwidth	180 deg. @64 GHz (E-plane) 82 deg. @64 GHz (H-plane)	
Polarization	Linear	
Max Tx Power	8 dBm	On chip output

The figure below depicts an antenna and its E and H planes:

- E-plane – plane XZ (green)
- H-plane – plane YZ (red)

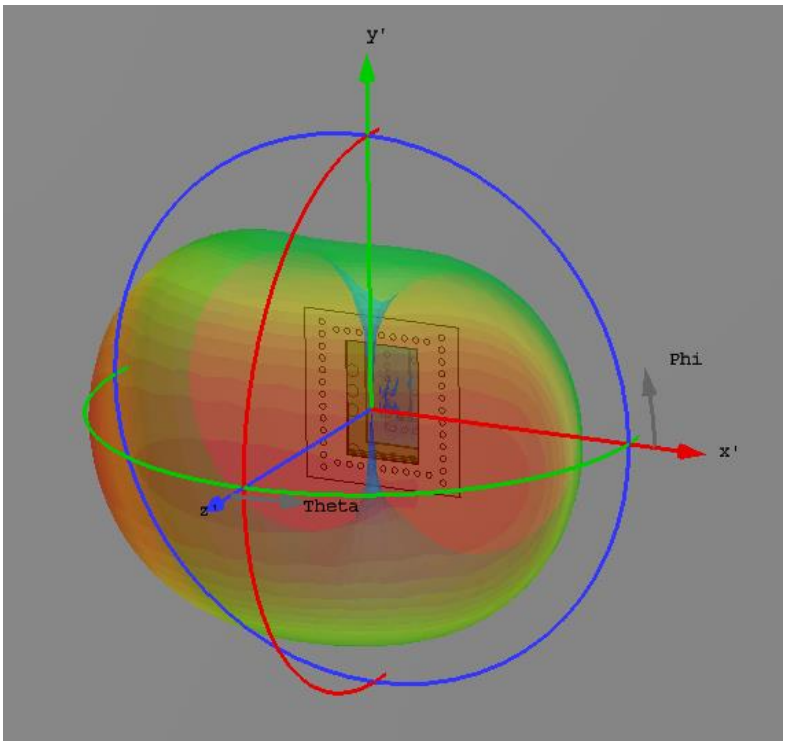


Figure 5: vTrig Antenna Pattern

2.3 Software

The vTrig system software performs signal processing and generation of the arena image.

The Graphical User Interface (GUI) provides images and blobs corresponding to the detected objects in the arena.

NOTE

The Graphical User Interface supports use of a single sensor only.

The following diagram describes the processing flow of the software.

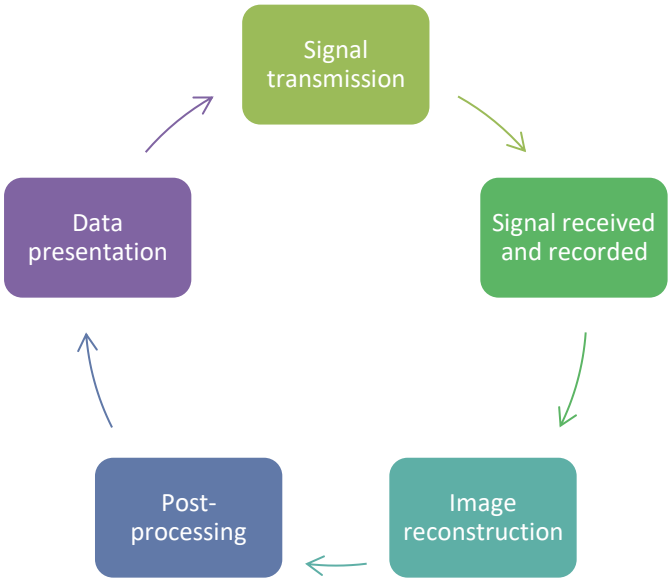


Figure 6: Processing Flow

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